Method: Hierarchical Bandit-Guided Risk-Sensitive Policy Optimization

This section formalizes the method presented in the supplied slides, using the current fixed-β SPO-RS formulation, hierarchical strategist–coder generation, adaptive hurdle-bandit allocation, and archive-based parent selection. :codex-file-citation{path="/Users/mohammad/Downloads/HQ-repotyt (6).pdf" purpose="source"}

1. Problem Formulation

Let x denote a scientific discovery task. A candidate-generating program y is executed by a task-specific environment,

$$a = \mathcal{E}_x(y),$$

where a is the resulting scientific construction. A deterministic evaluator assigns a scalar reward

$$R_x(a) \in \mathbb{R}.$$

The reward representation is normalized so that larger values are always preferred, even when the native scientific metric is minimized. Invalid programs receive a designated failure reward r_{fail}.

Discovery proceeds over iterations $t = 0, \ldots, T-1$. At iteration t, the system maintains:

$$\mathcal{A}_t = \{s_1, \ldots, s_{N_t}\},$$

an archive of previously evaluated scientific states. A state contains its program, resulting construction, normalized reward, native metric, creation step, and ancestral lineage.

The trainable coder is a conditional language-model policy

$$\pi_{\theta_t}(y \mid x, p, z),$$

where $p \in \mathcal{A}_t$ is a selected parent and z is a high-level strategy. The strategist is a separate frozen policy

$$q_\phi(z \mid x, p, \mathcal{H}),$$

where $\mathcal{H}$ contains earlier strategies generated for the same parent. Only θ is updated by reinforcement learning; ϕ remains fixed.

The discovery objective is to find the best valid construction encountered during the search:

$$a^\star = \arg\max_{a \in \mathcal{V}_{0:T}} R_x(a),$$

where $\mathcal{V}_{0:T}$ is the set of all valid evaluated constructions, including candidates later removed from the bounded archive.

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2. Hierarchical Strategist–Coder Policy

Direct program generation forces one model to perform two different functions simultaneously:

1. invent a scientifically meaningful direction; and
2. translate that direction into an executable program.

The proposed hierarchy factorizes these functions. For each selected parent p, the strategist generates S strategies sequentially:

$$z_{p,k} \sim q_\phi(z \mid x, p, z_{p,1:k-1}), \qquad k = 1, \ldots, S.$$

Thus,

$$q_\phi(z_{p,1:S} \mid x, p) = \prod_{k=1}^{S} q_\phi(z_{p,k} \mid x, p, z_{p,1:k-1}).$$

Earlier strategies are supplied only within the same parent chain. The strategist is instructed to produce a materially distinct proposal rather than a superficial variation. Different parent chains are conditionally independent and can be processed in parallel.

The coder then samples programs separately for each strategy:

$$y_{p,k,j} \sim \pi_{\theta_t}\left(y \mid x, p, z_{p,k}\right).$$

The coder receives the original task, the selected parent, and one extracted strategy. It does not receive the strategist's hidden reasoning or the other strategies. The strategy is treated as guidance: the coder may correct or adapt it when it conflicts with the task, mathematical validity, or the parent construction.

The resulting hierarchical proposal distribution is

$$P_t(p, z, y) = \alpha_t(p)\, q_\phi(z \mid x, p, \mathcal{H}_p)\, \omega_t(z \mid p)\, \pi_{\theta_t}(y \mid x, p, z),$$

where:

- $\alpha_t(p)$ is induced by archive-level parent selection;
- q_ϕ generates high-level strategies;
- $\omega_t(z \mid p)$ is induced by the strategy-allocation bandit;
- π_{θ_t} generates programs.

No policy gradient is propagated through parent selection, the strategist, or the bandit. The coder is trained on the trajectories produced under the resulting adaptive prompt distribution.

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3. Archive-Level Parent Bandit

Parent selection is performed by a PUCT-style archive bandit. For each state s, define:

- $R(s)$: reward of the state itself;
- $n(s)$: number of rollouts allocated to s or any descendant of s;
- $m(s)$: best child reward observed from s;
- $T = \sum_s n_{\text{direct}}(s)$: total number of rollout expansions;
- $P(s)$: a rank-based archive prior.

The exploitation statistic is

$$Q(s) = \begin{cases} m(s), & n(s) > 0, \\ R(s), & n(s) = 0. \end{cases}$$

Using the best observed child rather than the mean aligns the archive bandit with best-candidate scientific discovery.

Let $N = |\mathcal{A}_t|$, and let $\text{rank}(s) \in \{0, \ldots, N-1\}$, with rank zero assigned to the highest-reward state. The rank prior is

$$P(s) = \frac{N - \text{rank}(s)}{\sum_{j=1}^{N} j}.$$

The reward scale is

$$\Delta_t = \max\left(\max_{s \notin \mathcal{S}_0} R(s) - \min_{s \notin \mathcal{S}_0} R(s), \delta\right),$$

where $\mathcal{S}_0$ is the seed set and $\delta > 0$ prevents a zero exploration scale.

The PUCT score is

$$U_t(s) = Q(s) + c_{\text{PUCT}}\, \Delta_t\, P(s)\, \frac{\sqrt{1+T}}{1+n(s)}.$$

A corresponding prior-free UCT form is

$$U_t^{\mathrm{UCT}}(s) = Q(s) + c_{\mathrm{PUCT}}\,\Delta_t \sqrt{\frac{\log(1+T)}{1+n(s)}}.$$

Parents are sorted by this score. During batch construction, selecting a state temporarily blocks that state, its ancestors, and its descendants. This lineage blocking discourages a single successful lineage from occupying the entire parent batch. If blocking prevents the batch from being filled, the remaining positions are filled by the highest-scoring unselected states.

When K rollouts are assigned to parent p,

$$n(p) \leftarrow n(p) + K$$

and the same increment is propagated to every ancestor of p. Hence, repeated exploration anywhere in a lineage reduces the future exploration bonus of that whole lineage.

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4. Sequential Strategy Generation

For every selected parent, the system constructs one strategy chain. Strategy k is generated only after strategies $1, \ldots, k-1$ for that parent have been accepted:

$$z_{p,k} \sim q_\phi(\cdot \mid x, p, z_{p,<k}).$$

The strategist is asked to supply:

- the scientific rationale;
- a precise mathematical specification;
- the algorithmic procedure;
- consequential search choices;
- numerical and feasibility safeguards;
- the required result contract.

The strategist is also told that the downstream coder is a one-shot model. Accordingly, the plan must express evaluation, repair, adaptation, and fallback procedures as operations for the generated program to implement. It must not instruct the coder to run preliminary tests, wait for feedback, or revise the answer after execution.

This design treats a strategy as an externally generated, high-level latent action. Because the strategist is frozen, its proposal distribution does not drift during coder training. Diversity is encouraged through explicit dependence on prior strategies, while coder autonomy prevents an overly prescriptive plan from becoming a hard constraint.

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5. Two-Stage Strategy Rollout Allocation

Each strategy is treated as a bandit arm whose reward distribution is induced by stochastic coder generation and program execution.

Let:

- S be the number of strategies for a parent;
- M be the nominal number of programs per strategy;
- H be the pilot count per strategy;
- $B = S(M - H)$ be the remaining phase-two budget.

The total parent budget remains fixed:

$$N_{\mathrm{parent}} = SM = SH + B.$$

First, exactly H pilot programs are sampled from every strategy. The pilot outcomes are evaluated before phase-two counts are assigned.

For strategy k, let $R_{k,j}$ be pilot reward, $V_{k,j} \in \{0, 1\}$ indicate evaluator validity, and R_p be the saved parent reward. Define parent-relative improvement

$$Z_{k,j} = \begin{cases} R_{k,j} - R_p, & V_{k,j} = 1 \text{ and } R_{k,j} > R_p, \\ 0, & \text{otherwise.} \end{cases}$$

This creates a hurdle model with two components:

1. whether the strategy produces an improvement;
2. the improvement magnitude conditional on producing one.

Invalid programs, valid regressions, and unchanged candidates all belong to the zero component.

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6. Posterior Improvement Probability

Let

$$m_k = \sum_{j=1}^{H} \mathbf{1}\{Z_{k,j} > 0\}$$

be the number of positive improvements. With a uniform Beta prior, the improvement probability satisfies

$$p_k \mid \mathcal{D}_k \sim \text{Beta}\left(1 + m_k,\, 1 + H - m_k\right).$$

Its posterior mean is

$$\mathbb{E}[p_k \mid \mathcal{D}_k] = \frac{m_k + 1}{H + 2}.$$

This probability concerns strict improvement over the selected parent. It is different from mere program validity.

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7. Conditional Positive-Gain Model

The positive observations for arm k are

$$\mathcal{Z}_k^+ = \{Z_{k,j} : Z_{k,j} > 0\}.$$

All positive improvements from the strategies of the same parent are pooled to construct a weak common prior. Let

$$z_{\max} = \max_{k,j} Z_{k,j}, \qquad \widetilde{Z}_{k,j} = \frac{Z_{k,j}}{z_{\max}}.$$

This normalization makes the allocation invariant to the absolute reward scale.

A reflected Gaussian kernel gives a distribution supported on nonnegative gains. For kernel center $c > 0$ and bandwidth $h > 0$, its CDF is

$$K_h(z \mid c) = \Phi\left(\frac{z - c}{h}\right) + \Phi\left(\frac{z + c}{h}\right) - 1, \qquad z \geq 0.$$

The shared bandwidth is based only on positive pilot gains:

$$h = 0.9\,\widehat{\sigma}\, n_+^{-1/5},$$

where n_+ is the pooled positive count and

$$\widehat{\sigma} = \min\left(s,\, \frac{\text{IQR}}{1.349}\right)$$

when both scales are positive. A positive mean-based scale is used when the observed gains have no identifiable spread.

For arm k, construct mixture components consisting of:

- one zero-outcome component;
- one pooled positive-gain component;
- one component for every positive pilot observation of arm k.

Let the corresponding mixture weights be

$$w_k \sim \text{Dirichlet}(\boldsymbol{\alpha}_k),$$

with concentrations

$$\boldsymbol{\alpha}_k = \left(H - m_k + 1,\, 1,\, \underbrace{1, \ldots, 1}_{m_k} \right).$$

The expected total weight on the positive components is therefore

$$\frac{m_k + 1}{H + 2},$$

which exactly agrees with the Beta hurdle posterior mean.

Conditional on mixture weights, the predictive CDF is

$$F_k(z \mid w_k) = \sum_q w_{k,q} F_{k,q}(z).$$

Because several future samples from one strategy share the same uncertain arm distribution, the required quantity is the posterior CDF moment

$$M_{k,a}(z) = \mathbb{E}_{w_k}\left[F_k(z \mid w_k)^a \right],$$

rather than $\mathbb{E}[F_k(z)]^a$.

For Dirichlet parameters α_q, $A = \sum_q \alpha_q$, and component CDF values $F_q(z)$, this moment is

$$M_{k,a}(z) = \frac{1}{(A)_a} \sum_{\substack{n_1 + \cdots + n_Q = a \\ n_q \geq 0}} \frac{a!}{\prod_q n_q!} \prod_q (\alpha_q)_{n_q} F_q(z)^{n_q},$$

where $(u)_n$ is the rising factorial. These moments can be evaluated through a stable complete-homogeneous-polynomial recurrence.

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8. Fixed-Budget Expected-Best Allocation

Let a_k be the number of phase-two rollouts assigned to strategy k, with

$$a_k \in \mathbb{Z}_{\geq 0}, \qquad \sum_{k=1}^{S} a_k = B.$$

Let the current incumbent be the best of the parent and all valid pilots:

$$R_{\text{inc}} = \max\left(R_p,\, \max_{k,j : V_{k,j} = 1} R_{k,j} \right).$$

Equivalently, the current best positive gain is

$$z_{\text{inc}} = R_{\text{inc}} - R_p.$$

Under independent arm posteriors, the posterior expected improvement of allocation $\mathbf{a}$ is

$$\text{EI}(\mathbf{a}) = \int_{z_{\text{inc}}}^{\infty} \left[1 - \prod_{k=1}^{S} M_{k,a_k}(z)\right] dz.$$

The phase-two budget is allocated greedily. Starting from $\mathbf{a} = \mathbf{0}$, each unit is assigned to

$$k^\star = \arg\max_k \left[\text{EI}(\mathbf{a} + \mathbf{e}_k) - \text{EI}(\mathbf{a})\right].$$

The marginal gain can be written as

$$\Delta_k(\mathbf{a}) = \int_{z_{\text{inc}}}^{\infty} \left[\prod_{r=1}^{S} M_{r,a_r}(z)\right] \left[1 - \frac{M_{k,a_k+1}(z)}{M_{k,a_k}(z)}\right] dz.$$

The integral is evaluated deterministically by Gaussian quadrature. After gain normalization, the numerical interval extends from the observed best gain to twelve kernel bandwidths beyond it, where the remaining Gaussian tail is negligible.

This allocation directly targets the expected best final result, rather than the average phase-two reward. It therefore rewards both:

- a high posterior chance of improvement; and
- a sufficiently heavy conditional upper tail.

All phase-two allocations are computed from the completed pilots. There is no additional allocation barrier after every phase-two rollout.

If no strategy improves the saved parent during the pilot stage, the positive-gain scale is unidentifiable. The remaining budget is then divided equally. Failure severity and validity rates are deliberately not used to invent a preference about unobserved improvement magnitudes.

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9. Group-Relative Entropic Advantage

The framework supports a group-relative entropic advantage as a baseline method. For K rollouts from one parent, let the rewards be $r_1, \ldots, r_K$. Define

$$q_\beta(i) = \frac{\exp(\beta r_i)}{\sum_{j=1}^{K} \exp(\beta r_j)}.$$

The inverse temperature β is selected so that

$$D_{\text{KL}}\left(q_\beta \parallel \text{Uniform}(K)\right) = \gamma,$$

subject to a finite upper bound. This makes the concentration of the exponential weights comparable across groups.

Using a numerically stable shift $r_{\max} = \max_j r_j$, define

$$e_i = \exp\left(\beta(r_i - r_{\max})\right),$$

and the leave-one-out baseline

$$B_{-i} = \frac{1}{K-1} \sum_{j \neq i} e_j.$$

The implemented group-relative advantage is

$$A_i^{\mathrm{ENT}} = \frac{e_i}{B_{-i}} - 1.$$

Thus a rollout is reinforced when its exponentiated reward exceeds the average exponentiated reward of its peers. An all-equal group receives

$$A_i^{\mathrm{ENT}} = 0$$

for every rollout.

The code-level group-relative mode uses the expression above without a $1/\beta$ prefactor. This scaling is distinct from the SPO-RS advantage defined next.

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10. SPO-RS Risk-Sensitive Objective

SPO-RS, called SPO^2 in the slides, combines a persistent single-stream baseline with an exponential risk-sensitive reward transform.

For $\beta > 0$, define the entropic objective over the active mixture of parents and strategies:

$$J_\beta(\theta) = \frac{1}{\beta} \log \mathbb{E}_{c\sim\nu_t,\, y\sim\pi_\theta(\cdot|c)} \left[e^{\beta R(y)} \right],$$

where $c = (x, p, z)$ denotes the coder context and ν_t is the context distribution induced by parent selection, strategy generation, and bandit allocation.

The cumulant expansion is

$$J_\beta(\theta) = \mathbb{E}[R] + \frac{\beta}{2} \operatorname{Var}(R) + O(\beta^2).$$

Consequently, positive β favors upper-tail outcomes in addition to mean reward. As $\beta \to 0$, the objective approaches ordinary expected reward.

The current method uses one fixed β throughout training. This prevents late-stage near-tied reward groups from driving an adaptive inverse temperature toward an extreme value and producing unstable, nearly winner-take-all gradients.

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11. Frozen Run-Wide SPO-RS Baseline

Define the transformed reward

$$G(r) = e^{\beta r}.$$

Before iteration t, the system maintains one positive run-wide baseline

$$v_{t-1} > 0.$$

Every rollout generated at iteration t, across all selected parents and all strategies, receives advantage

$$A_{t,i}^{\text{SPO-RS}} = \frac{1}{\beta} \left(\frac{G(r_{t,i})}{v_{t-1}} - 1 \right).$$

The same frozen v_{t-1} is used for the entire iteration. Current rewards are not incorporated into the baseline until all current advantages have been formed.

This differs from group-relative normalization in several ways:

- one rollout is sufficient to obtain an advantage;
- tied rollouts can still receive nonzero learning signals;
- the same reward receives the same sign relative to the historical baseline;

- different parents and strategy allocations share a common reward scale;
- one rollout cannot change the advantage assigned to another rollout in the same iteration.

For a frozen action-independent $v > 0$,

$$\begin{aligned}\mathbb{E}\left[A^{\text{SPO-RS}}\nabla_\theta \log \pi_\theta(y \mid c)\right] &= \frac{1}{\beta v}\mathbb{E}\left[e^{\beta R(y)}\nabla_\theta \log \pi_\theta(y \mid c)\right] \\ &= \frac{M_\beta(\theta)}{v}\nabla_\theta J_\beta(\theta),\end{aligned}$$

where

$$M_\beta(\theta) = \mathbb{E}[e^{\beta R(y)}].$$

The subtraction of 1 contributes no expected score-function gradient because

$$\mathbb{E}_{y\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(y \mid c)] = 0.$$

Therefore, before clipping and finite-sample effects, an imperfect positive historical baseline changes the gradient magnitude but not its ascent direction for the entropic objective.

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12. Policy-Divergence-Adaptive Historical Memory

The baseline must retain information when the coder changes slowly and forget stale evidence when the coder changes substantially.

Let $\mu_t = \pi_{\theta_t}$ be the policy that generated iteration t, and let μ_{t-1} be the preceding rollout policy. For context c, current trajectories are sampled from μ_t. The trajectory-level forward-KL estimator is

$$\widehat{D}_{t,c} = \frac{1}{N_c}\sum_{i\in c}\sum_{\ell=1}^{L_i}\left[\log \mu_t(y_{i,\ell} \mid y_{i,<\ell}, c) - \log \mu_{t-1}(y_{i,\ell} \mid y_{i,<\ell}, c)\right].$$

Contexts are weighted equally:

$$D_t = \max\left(0,\ \frac{1}{|\mathcal{C}_t|}\sum_{c\in\mathcal{C}_t}\widehat{D}_{t,c}\right).$$

The nonnegative clamp removes small negative Monte Carlo deviations from the nonnegative population KL.

The retention factor is

$$\rho_t = \text{clip}\left(2^{-D_t/D_{1/2}}, \rho_{\min}, \rho_{\max}\right),$$

where $D_{1/2} > 0$ is the divergence at which unconstrained historical retention is halved. Because $\rho_{\max} < 1$, the tracker eventually forgets old evidence even when the policy is nearly unchanged. If the divergence cannot be measured, minimum retention is used.

Let C_{t-1} be the tracker's effective historical count and let M_t be the number of current rollouts. Define

$$\overline{G}_t = \frac{1}{M_t}\sum_{i=1}^{M_t} G(r_{t,i}).$$

After all advantages have been frozen, update

$$C_t = \rho_t C_{t-1} + M_t,$$

$$v_t = \frac{\rho_t C_{t-1} v_{t-1} + \sum_{i=1}^{M_t} G(r_{t,i})}{C_t}.$$

Equivalently,

$$v_t = (1 - \eta_t) v_{t-1} + \eta_t \overline{G}_t, \qquad \eta_t = \frac{M_t}{C_t}.$$

The baseline is initialized before the first rollout iteration from verified seed-state rewards. This enables the first iteration to produce valid advantages without allowing its own rewards to define their normalization.

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13. SPO-RS Clipped Policy Objective

For a generated trajectory $y_i = (y_{i,1}, \ldots, y_{i,L_i})$, store token log-probabilities under the frozen behavior policy

$$\mu_t = \pi_{\theta_t}.$$

During optimization, define the token importance ratio

$$r_{i,\ell}(\theta) = \exp\left[\log \pi_\theta(y_{i,\ell} \mid y_{i,<\ell}, c_i) - \log \mu_t(y_{i,\ell} \mid y_{i,<\ell}, c_i)\right].$$

Let

$$\bar{r}_{i,\ell}(\theta) = \mathrm{clip}\left(r_{i,\ell}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}\right).$$

The asymmetric clipped loss for one trajectory is

$$\mathcal{L}_i^{\text{SPO-RS}}(\theta) = -\frac{1}{L_i} \sum_{\ell=1}^{L_i} \min\left[r_{i,\ell}(\theta) A_i,\ \bar{r}_{i,\ell}(\theta) A_i\right].$$

The complete loss is an equal average over all usable trajectories:

$$\mathcal{L}^{\text{SPO-RS}}(\theta) = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathcal{L}_i^{\text{SPO-RS}}(\theta).$$

Pilot and phase-two rollouts are treated identically during policy optimization. Because the bandit assigns more trajectories to promising strategies, those strategies intentionally contribute more observations to the coder update.

Tied-reward groups are retained because their advantages are determined by the historical baseline rather than within-group differences. Invalid rollouts also remain part of the reward distribution and can provide negative learning signals through the failure reward.

The behavior log-probabilities and advantages are frozen during optimization. The implementation normally performs a small fixed number of update epochs and clips the trainable gradient norm. A separate reference-policy KL penalty is disabled for SPO-RS; stability is supplied by the frozen behavior policy, clipped ratios, and divergence-adaptive tracker memory.

For architectures requiring a trajectory-stable ratio, the token ratios can be replaced by the geometric-mean sequence ratio

$$r_i^{\text{seq}}(\theta) = \exp\left[\frac{1}{L_i} \sum_{\ell=1}^{L_i} \left(\log \pi_\theta(y_{i,\ell} \mid \cdot) - \log \mu_t(y_{i,\ell} \mid \cdot)\right)\right],$$

and the same scalar is used throughout the trajectory.

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14. Entropic-Mode Policy Training

For the group-relative entropic mode, let π_0 denote a frozen base policy and define

$$d_{i,\ell} = \log \pi_\theta(y_{i,\ell} \mid \cdot) - \log \pi_0(y_{i,\ell} \mid \cdot),$$

$$\bar{d}_i = \frac{1}{L_i} \sum_{\ell=1}^{L_i} d_{i,\ell}.$$

A centered reference-policy correction is

$$A_{i,\ell}^{\mathrm{REF}} = \lambda \left(\bar{d}_i - d_{i,\ell} \right).$$

The effective token advantage is

$$A_{i,\ell}^{\mathrm{eff}} = A_i^{\mathrm{ENT}} + A_{i,\ell}^{\mathrm{REF}}.$$

With detached behavior importance weights

$$w_{i,\ell} = \exp\left[\log \pi_\theta(y_{i,\ell} \mid \cdot) - \log \mu_t(y_{i,\ell} \mid \cdot) \right],$$

the policy-gradient loss is

$$\mathcal{L}_i^{\mathrm{ENT}} = -\frac{1}{L_i} \sum_{\ell=1}^{L_i} \mathrm{stopgrad}\left(w_{i,\ell} A_{i,\ell}^{\mathrm{eff}} \right) \log \pi_\theta(y_{i,\ell} \mid \cdot).$$

Constant-reward groups are excluded because their entropic advantages are identically zero.

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15. Candidate Admission and Archive Update

Every executed program produces a durable candidate record, irrespective of whether it ultimately survives archive pruning. Valid candidates enter the archive-admission procedure.

First, exact program duplicates are removed. Then, for each parent–strategy pair, only the top R_{strategy} new children are admitted. This prevents a strategy that generated many similar high-scoring programs from monopolizing the archive.

Next, only the top K_{child} children of each parent are retained. Finally, the non-seed archive is globally capped by reward, while all seed states remain available.

The archive update therefore applies the composition

$$\begin{aligned} \text{evaluated candidates} &\to \text{exact deduplication} \to \text{top per parent–strategy} \\ &\to \text{top per parent} \to \text{global cap}. \end{aligned}$$

The parent exploitation statistic is updated using the best valid child before pruning:

$$m(p) \leftarrow \max\left(m(p), \max_{s \in \mathrm{child}(p)} R(s) \right).$$

The best native-metric result ever evaluated is maintained separately from the bounded archive. Consequently, deduplication or archive pruning cannot erase the run's best scientific result.

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16. Complete Training Procedure

At iteration t, the complete procedure is:

1. Freeze the pre-step SPO-RS baseline v_{t-1}.

2. Select a batch of parent states with archive PUCT and lineage blocking.
3. Record the rollout budget against each selected parent and its ancestors.
4. For every parent, generate a sequential chain of S materially different strategies using the frozen strategist.
5. Construct one coder prompt for each parent–strategy pair.
6. Generate H pilot programs independently from every strategy-conditioned coder distribution.
7. Execute and evaluate all pilot programs.
8. Fit the parent-specific hurdle posteriors from strict improvements over that parent.
9. Allocate the remaining fixed rollout budget by greedy posterior expected-best improvement.
10. Generate and evaluate the allocated phase-two programs.
11. Estimate the forward divergence between the current and previous rollout policies on the current parent contexts.
12. Transform every reward using $G(r) = e^{\beta r}$.
13. Compute every SPO-RS advantage using the same frozen v_{t-1}.
14. Update the historical tracker only after all current advantages have been formed.
15. Save behavior log-probabilities and complete rollout records.
16. Admit valid children through the strategy and parent archive filters.
17. Optimize the coder adapter using the clipped SPO-RS objective.
18. Freeze the resulting coder policy as the behavior policy for the next iteration.
19. Preserve the globally best valid scientific result independently of archive membership.

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17. Exploration and Collapse-Control Mechanisms

The method uses several complementary exploration controls.

The strategist introduces high-level semantic variation before code generation. Sequential conditioning discourages duplicate plans, while coder autonomy allows incorrect or overly restrictive plans to be repaired.

The hurdle bandit reallocates expensive program executions toward strategies with evidence of useful upper-tail behavior. It avoids spending the full budget uniformly while preserving exploration through equal pilots and posterior uncertainty.

Archive PUCT selects promising scientific lineages while retaining an exploration bonus for underexpanded states. Lineage blocking prevents one branch from occupying an entire parent batch.

Per-strategy and per-parent archive filters preserve multiple strategic modes after evaluation. Exact construction and program records allow semantic progress to be distinguished from trivial parent copying.

Finally, unrestricted policy sampling retains the model's full token distribution:

$$\tau = 1, \qquad p_{\text{top}} = 1, \qquad k_{\text{top}} = 0, \qquad p_{\min} = 0,$$

with neutral repetition penalties. Exploration is therefore controlled by learned policy probabilities, hierarchical conditioning, and bandit allocation rather than additional decoding truncation.

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18. Theoretical Interpretation

The method separates exploration into three timescales.

The archive bandit chooses *where in the scientific lineage to search.* Its maximum-child exploitation statistic is appropriate for best-result discovery.

The strategist chooses *which scientific direction to investigate.* Its frozen distribution supplies semantic modes without adding a second moving RL policy.

The strategy bandit chooses *how much execution budget each direction receives.* Its hurdle posterior distinguishes the frequency of improvement from the magnitude of successful improvements.

SPO-RS determines *how evaluated programs update the coder.* Its fixed exponential utility prioritizes rare high rewards, while its persistent baseline avoids dependence on noisy local group statistics.

This separation prevents one mechanism from carrying all responsibilities. The strategist need not predict exact executable success; the bandit empirically measures that success. The bandit need not learn token-level behavior; the coder update performs that role. The coder need not invent every high-level direction internally; the strategist supplies a structured proposal.

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19. Scope and Limitations

The method does not guarantee a global scientific optimum. The archive score, strategist plans, hurdle posterior, and language-model policy are approximate search mechanisms.

The hurdle model assumes that positive-gain behavior can be represented by a smoothed mixture of observed and pooled positive gains. Its tail beyond observed improvements is a modeling assumption rather than a distribution-free guarantee.

The clipped PPO objective deliberately introduces bias to constrain policy movement. The exact risk-sensitive gradient equivalence applies before clipping, finite-sample approximation, and adaptive context selection.

The coder is trained under the bandit-induced strategy distribution. No inverse-propensity correction is applied to reconstruct a uniform distribution over strategies. The resulting emphasis on empirically promising strategies is intentional.

The global SPO-RS baseline aggregates multiple parents and strategy contexts. It provides consistent run-wide normalization but does not estimate a separate conditional value for every parent. Policy-divergence-adaptive forgetting limits the influence of stale evidence when the coder distribution changes.

The complete method is therefore a hierarchical, fixed-budget discovery procedure whose components are aligned with the final objective: locating rare, valid, high-reward scientific constructions while continuing to train the program-generating policy.